Escuela Técnica Superior de Ingenieros Informáticos

MSc. IN DATA SCIENCE

Máster Universitario en Ciencia de Datos

Practical application 3: unsupervised classification

Álvaro García Barragán

Escuela Técnica Superior de Ingenieros Informáticos, Universidad Politécnica de Madrid

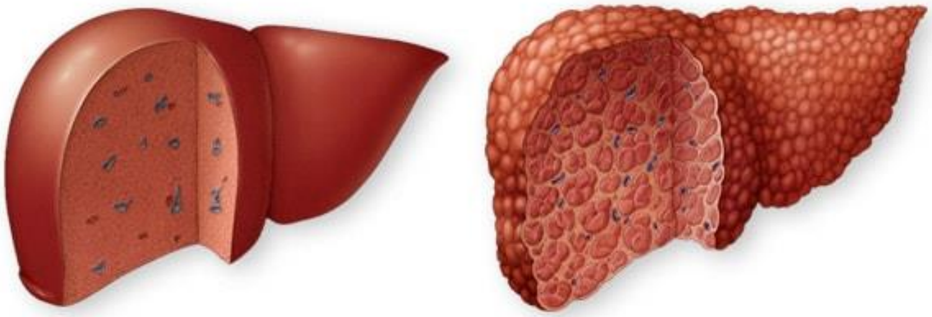
13^{th}December, 2023

Index

1. Introduction
2. Problem description
3. Methodology
4. Experiments & Results
5. Discussion & Conclusion

Introduction

Liver cirrhosis is a late stage of scarring (fibrosis) of the liver caused by many forms of liver diseases and conditions, such as hepatitis and chronic alcoholism.



Introduction

The data provided are sourced from a **Mayo Clinic**, from a study on primary biliary cirrhosis (PBC) carried out from 1974 to 1984. All the patients have suffered this disease and are censored or death.

Problem Description

- The dataset after preprocessing, contains **312 instances**, each representing an individual patient and each one, includes **20 clinical features**.
- Preprocessing encode qualitative features.

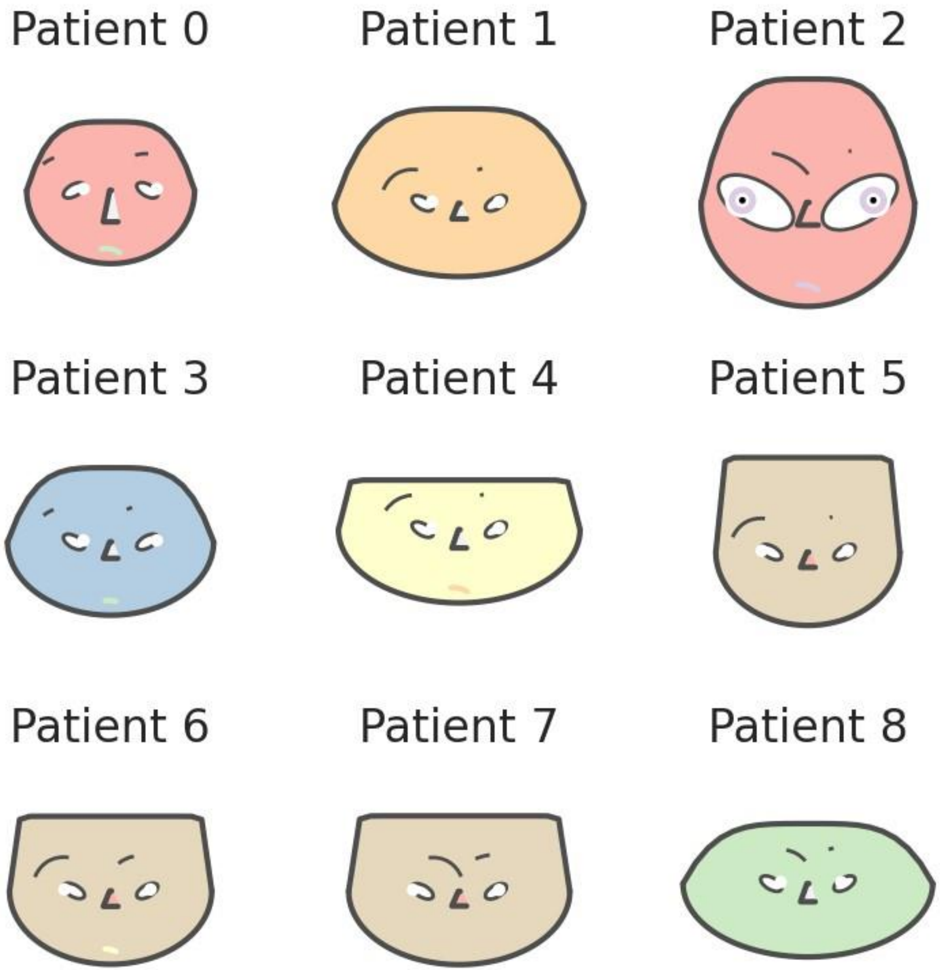
Table 1: Data set

Feature	Mean	Std	Min	25%	50%	75%	Max	Type
N_Days	2006.36	1123.28	41.00	1191.00	1839.50	2697.25	4556.00	continuous
Drug	0.49	0.50	0.00	0.00	0.00	1.00	1.00	binary (Y:0, N:1)*
Age	50.05	10.59	26.30	42.27	49.83	56.75	78.49	continuous
Sex	0.12	0.32	0.00	0.00	0.00	0.00	1.00	binary (F:0, M:1)
Ascites	0.08	0.27	0.00	0.00	0.00	0.00	1.00	binary (N:0, Y:1)
Hepatomegaly	0.51	0.50	0.00	0.00	1.00	1.00	1.00	binary (N:0, Y:1)
Spiders	0.29	0.45	0.00	0.00	0.00	1.00	1.00	binary (N:0, Y:1)
Bilirubin	3.26	4.53	0.30	0.80	1.35	3.42	28.00	continuous
Cholesterol	369.51	221.26	120.00	255.75	322.00	392.25	1775.00	continuous
Albumin	3.52	0.42	1.96	3.31	3.55	3.80	4.64	continuous
Copper	97.65	85.34	4.00	41.75	73.00	123.00	588.00	continuous
Alk_Phos	1982.66	2140.39	289.00	871.50	1259.00	1980.00	13862.40	continuous
SGOT	122.56	56.70	26.35	80.60	114.70	151.90	457.25	continuous
Tryglicerides	124.70	61.93	33.00	87.00	114.00	145.25	598.00	continuous
Platelets	261.94	94.99	62.00	200.00	258.50	322.00	563.00	continuous
Prothrombin	10.73	1.00	9.00	10.00	10.60	11.10	17.10	continuous
Stage	3.03	0.88	1.00	2.00	3.00	4.00	4.00	ordinal
Edema_N †	0.84	0.36	0.00	1.00	1.00	1.00	1.00	binary
Edema_Y †	0.06	0.25	0.00	0.00	0.00	0.00	1.00	binary
Edema_S †	0.09	0.29	0.00	0.00	0.00	0.00	1.00	binary

* Y=Drug=Dpenicillamine; N=Placebo;

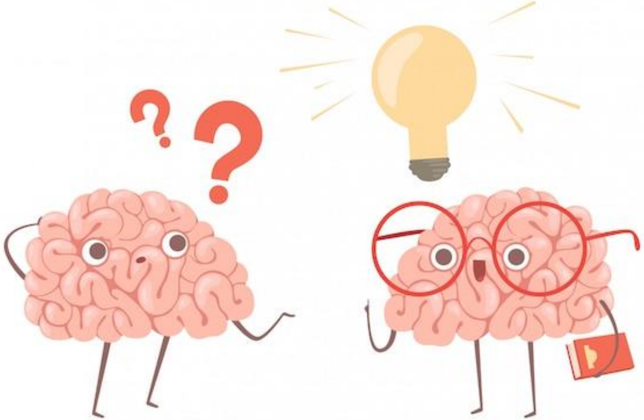
† Edema_N=No edema and no diuretic therapy for edema; Edema_Y=Edema despite diuretic therapy; Edema_S=Edema present without diuretics, or edema resolved by diuretics;

Chernoff faces



Problem Description

- What methods of clustering should be used only with **normalized data**?
- What if the best **k** for each method?
- What are the optimal criteria or **thresholds** in Hierarchical clustering?

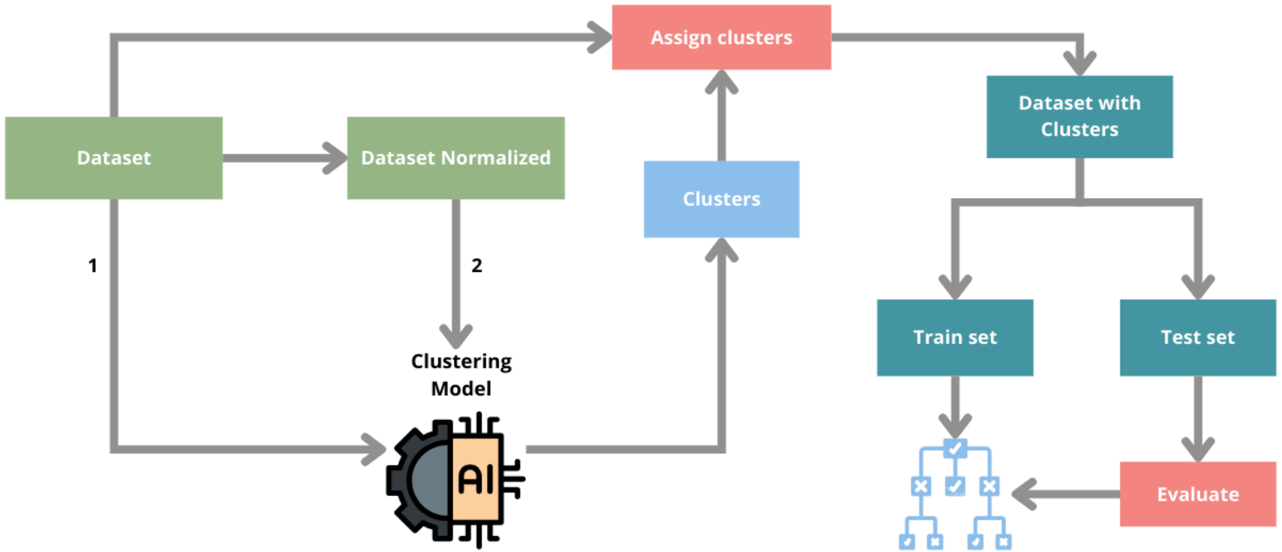


Approach

- Try to **adapt clustering to the business**, meaning that the configuration of k and the thresholds are manually adjusted until the results are interpretable. An expert in the domain or the client is needed.
- **Dimensionality reduction (PCA)** technique is required for visualizing the clusters in 2D space.
- To interpret the clustering, a **Decision Tree classifier** is trained to summarize the features of each cluster (max_depth=6, to be more interpretable)

Methodology

- **Dataset** is used to train the Decision Tree classifier model and clustering models that utilize Mahalanobis distance.
- **Dataset Normalized** is used is employed for PCA and other clustering methods that require normalization.



Experiments & Results

Table 2: Clustering methods and parameters

Method	Distance	threshold	k	F-score	Ref
Hierarchical Single	mahalanobis	5.6	16	0.9200	Figure 2
Hierarchical Complete	mahalanobis	12	5	0.7965	Figure 3
Hierarchical Complete	mahalanobis	10	10	0.6663	-
Hierarchical Centroid	euclidean	6	13	0.9015	-
Hierarchical Ward	euclidean	15	9	0.8792	Figure 4
Hierarchical Ward	euclidean	18	7	0.8940	-
Partitional KMeans	euclidean	-	3	0.8617	-
Partitional KMeans	euclidean	-	5	0.8243	Figure 6
Probabilistic Gaussian Mixture	-	-	3	1.0000	Figure 5
Probabilistic Gaussian Mixture	-	-	5	0.8901	-

Figure 2 (Single)

- 16 clusters, no stratified pattern, one cluster have many instances, other very few.

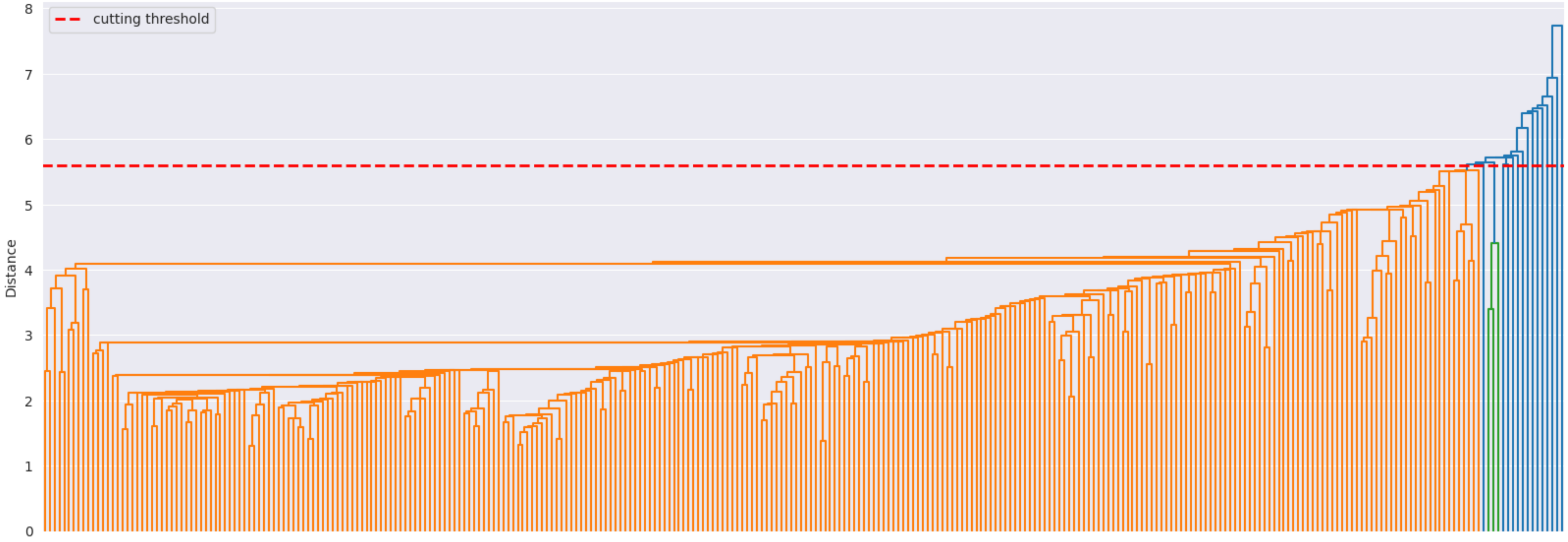


Figure 2 (Single)

- Don’t adjust well for the outliers.
- Almost every instance is in the same cluster.

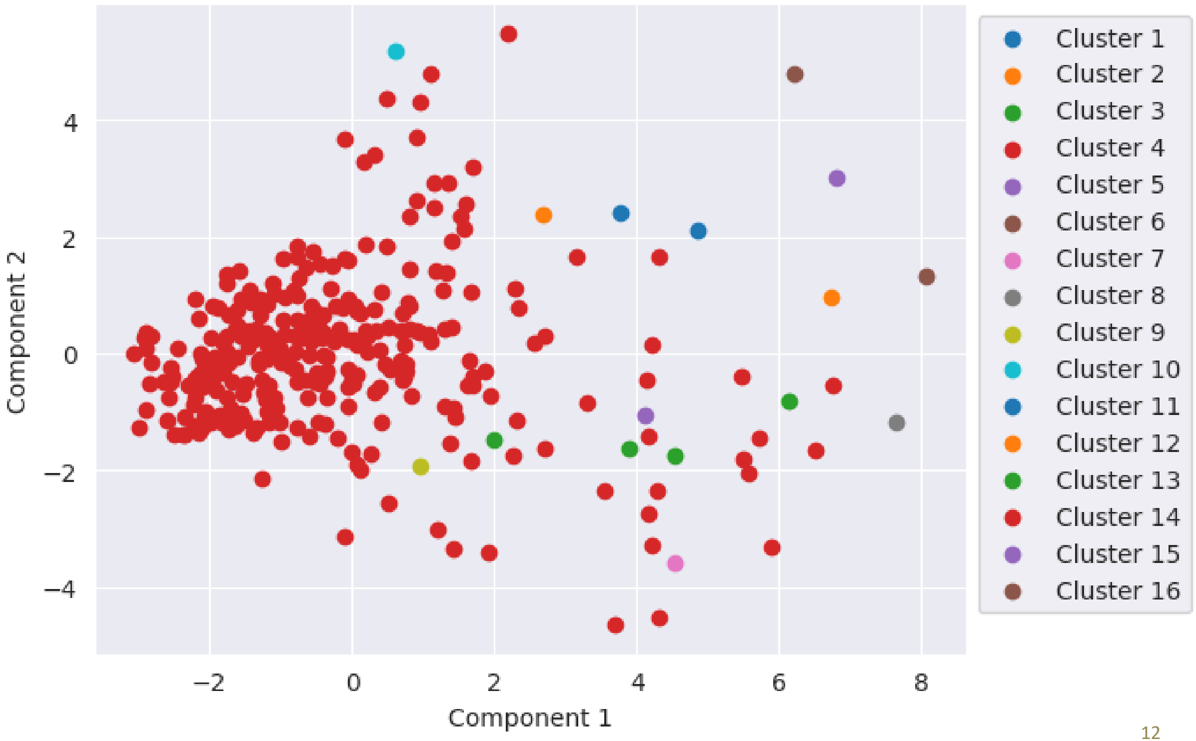


Figure 4 (Ward)

- Well balanced 9 clusters, more homogenous structure.

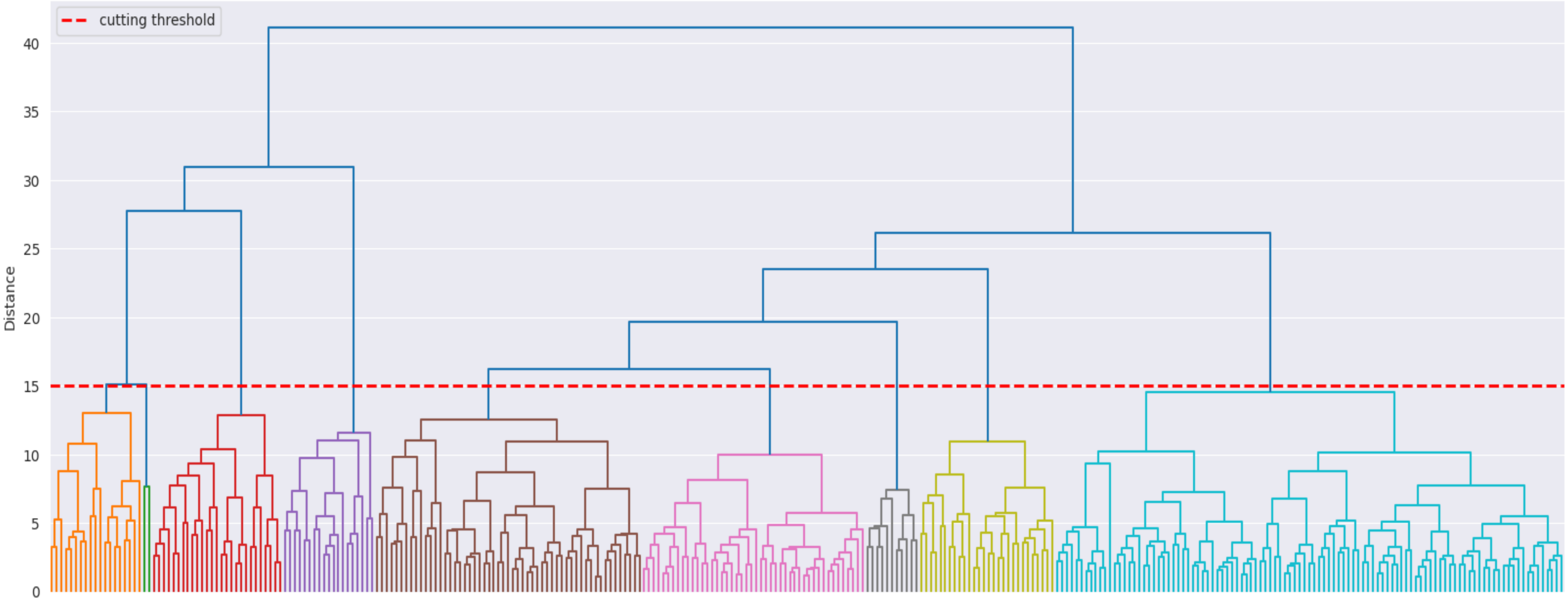


Figure 4 (Ward)

- Flexible clustering boundary margins.
- Each cluster with similar number of instances.

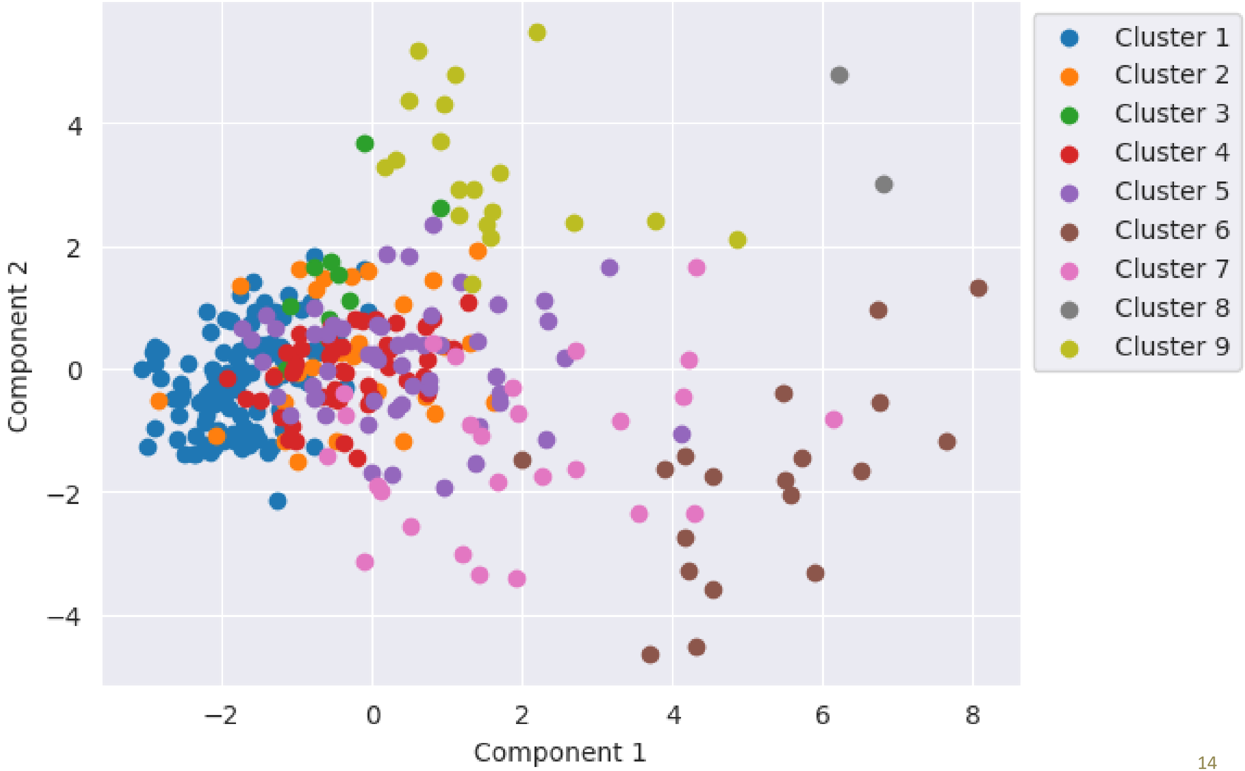


Figure 6 (KMeans)

- Rigid clustering boundary margins.
- Each cluster with similar number of instances.

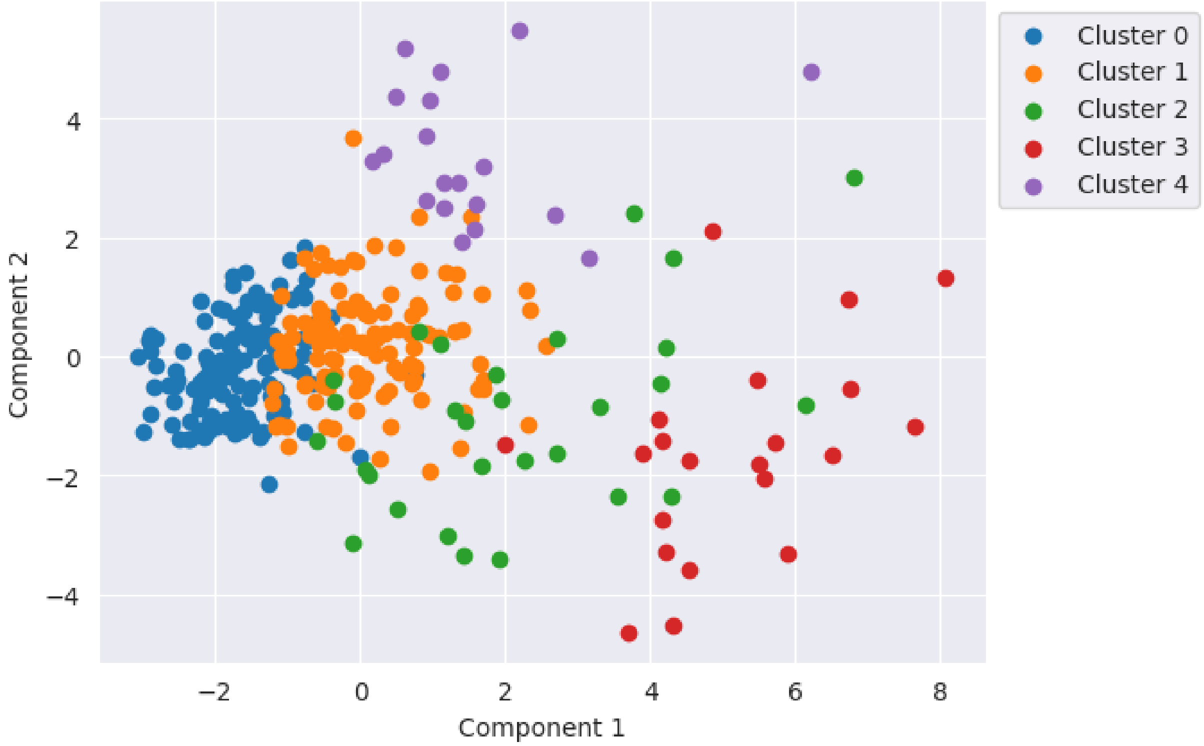


Figure 6 (Probabilistic Gaussian Mixture)

- Flexible clustering boundary margins.
- Each cluster with similar number of instances.

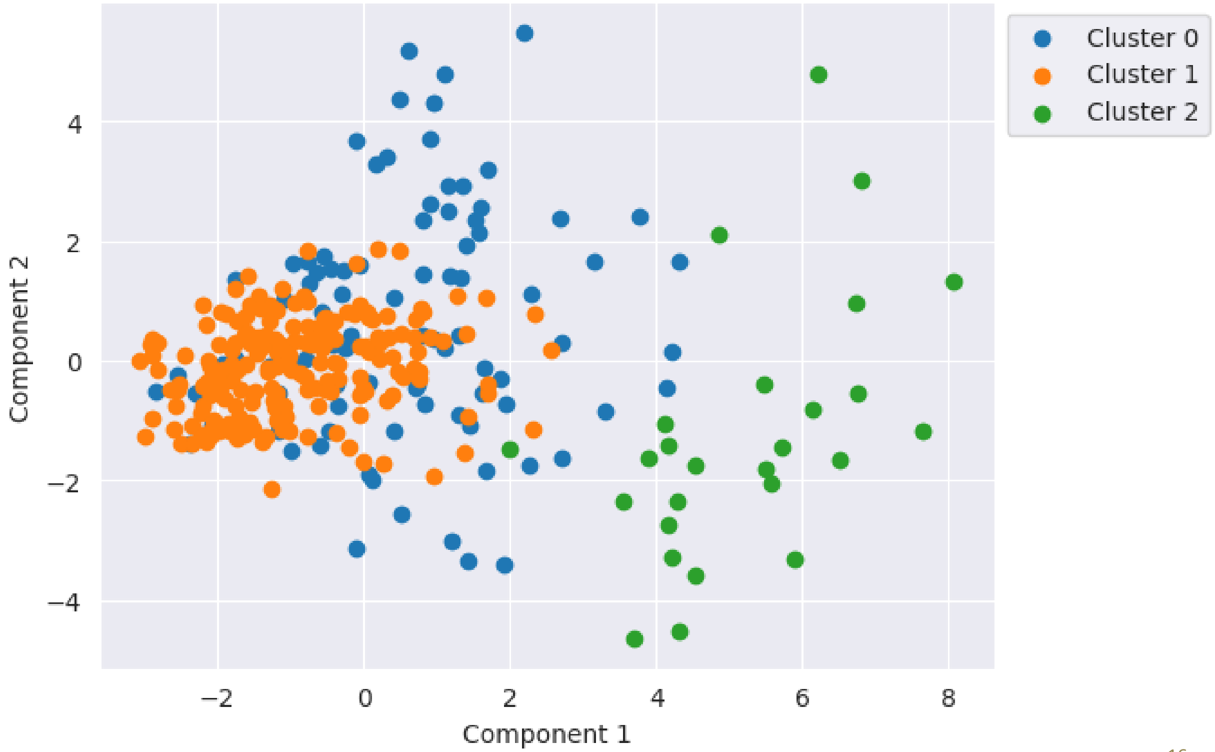


Figure 6 (Probabilistic Gaussian Mixture)

- Important to take in count *class: nº patients;* 0:92 ; 1:194 ; 2:26 (all data set)

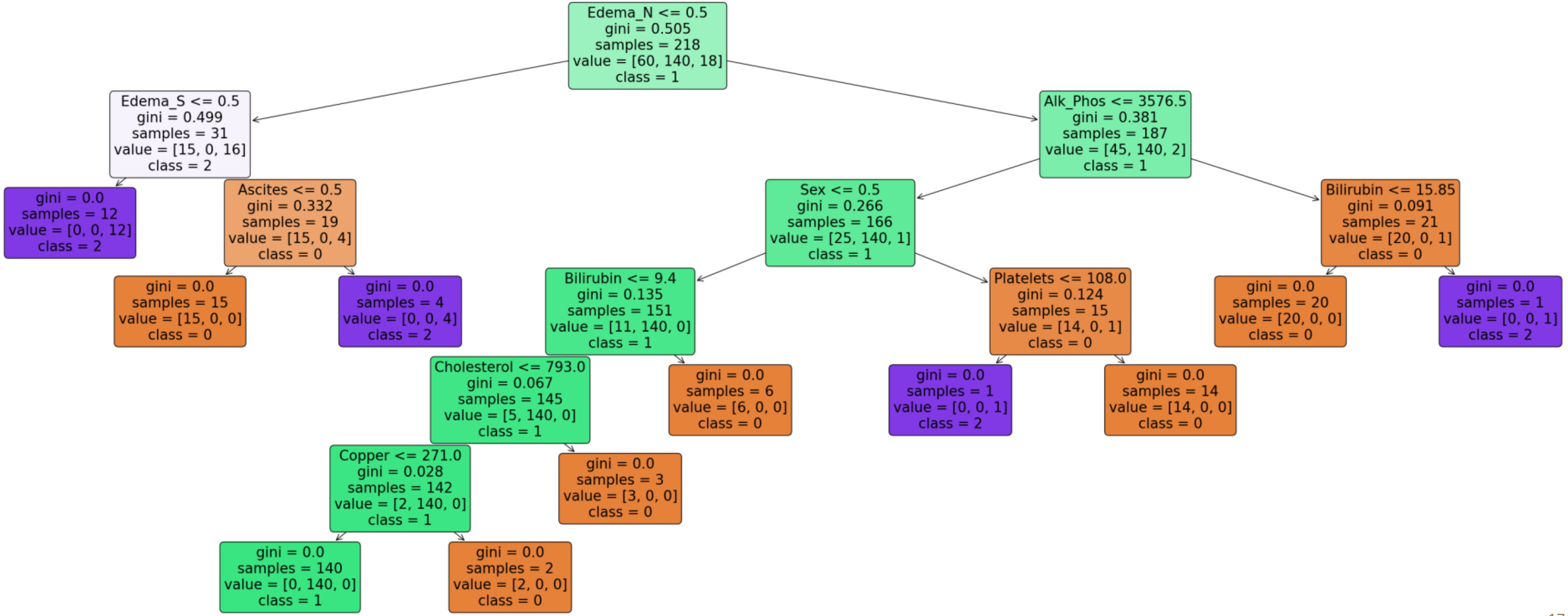


Figure 6 (Probabilistic Gaussian Mixture)

- *Cluster 0* (Class 0, 92 patients): **moderate-severity** biliary cirrhosis, as indicated by the absence of edema (Edema_N = 0) in 45 out of 60 patients. Only 15 out of 60 have Edeme_S = 1 and no ascites.
- *Cluster 1* (Class 1, 194 patients): **less severity** biliary cirrhosis, since all not having edema Edema_N = 0. The cluster is composed of females (Sex = 0).
- *Cluster 2* (Class 2, 26 patients): **advanced stage** of biliary cirrhosis (Edema_Y = 1). These patients may also have Ascites (4 out of 18 patients).

Discussion & Conclusion

- The smaller number of k and number of leaf nodes, the more interpretable clusters it are, but can be a loss of the capability of the model to select group the patients in a good cluster.
- Methods like Ward or Kmeans divide the clusters in a more homogenous manner, while others as Single are less stratified (balanced).
- Parsimony of decision tree classifiers with this dataset, the smaller number of classes the better F-score.
- Probabilistic clustering divide the patients in three distinct cohorts, related with an understanding of the disease.

Thanks!

Álvaro García Barragán
alvaro.gbarragan@alumnos.upm.es
Escuela Técnica Superior de Ingenieros Informáticos, Universidad Politécnica de Madrid